(*) Even at 2.5=β we are not able to achieve min tp. we can get min tp below 2.5=β.

(*) For improving delay we should use β should be around 2 or 1.9. But if your improving noise then β should be around 2.5.

11/9/25:

Constant Field scaling / constant voltage scaling : (check in kang book)

$$t_p = 0.69\, R_{eq}(C_{int} + C_{ext})$$

C_{int} ↑ internal or parasitic (intrinsic) — inside the transistor

C_{ext} ↑ external

$$t_p = 0.69\, R_{eq}\, C_{int}\left(1 + \frac{C_{ext}}{C_{int}}\right)$$

$$0.69\, R_{eq}\, C_{int} = t_{p_0}$$

S>1

$$R_{eq} = \frac{R_{ref}}{S} \qquad C_{int} = S \cdot C_{ref}$$

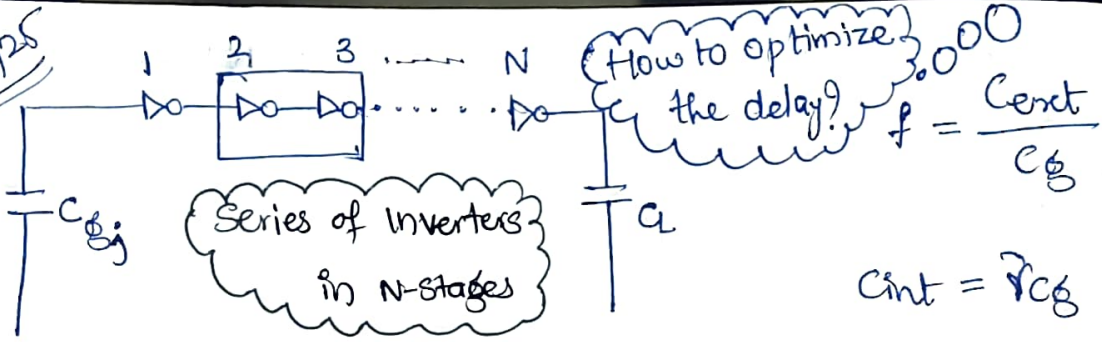
$$t_{p_0} = 0.69 \times \frac{R_{ref}}{S} \times S \cdot C_{ref}$$

t_{p_0} → doesn't change with scaling ← Intrinsic delay

$$\boxed{t_{p_0} = 0.69 \times R_{ref} \times C_{ref}}$$

t_p =

15/09/25



$f = \frac{C_{ext}}{C_g}$

$C_{int} = \gamma C_g$

$t_p = t_{p_0}\left(1 + \frac{C_{ext}}{C_{int}}\right)$

$$t_p = t_{p_0}\left(1 + \frac{f}{\gamma}\right)$$

Rabaey Book

$$t_{p,j} = t_{p_0}\left(1 + \frac{C_{g_{j+1}}}{\gamma C_{g_j}}\right)$$

Total delay, N-stage, $t_p = \sum_{j=1}^{N} t_{p_j}$

$$\frac{\partial t_p}{\partial C_{g,j}} = 0$$

after deriving, we get

$$f = \frac{C_{g,j+1}}{C_{g,j}} = \frac{C_{g,j}}{C_{g,j-1}}$$

Imp Fundamental Block

f ↑ fan out

⊙ what is called overall fan-out?

$F = f_1 f_2 f_3 f_4 \cdots$

$= \frac{C_j}{C_{j-1}} \cdot \frac{C_{j+1}}{C_j}$

$F = f^N$

$$f = \sqrt[N]{F}$$

$$t_p = N t_{p_0}\left(1 + \frac{\sqrt[N]{F}}{\gamma}\right)$$

It says that each stage provides same amount of delay.

How to get delay in optimized way in 3-stage, 4-stage, ?..

⊙ To take get equal delay in all stages → we need constant fan out

overall $f = \sqrt[N]{\frac{C_L}{C_{g,1}}} = \sqrt[N]{F}$

F = overall f

Do derivative in terms N so that we get the N value.

$$\frac{\partial t_p}{\partial N} = 0$$

$$\gamma + \sqrt[N]{F} - \frac{\sqrt[N]{F}\ln F}{N} = 0$$

0 → Self-loading is ignored. Short-channel effect & other things are ignored.

1 → opposite of 0, i.e., not ignored. Short-channel effect & other things

⊙ $\gamma = 0 \rightarrow N\sqrt[N]{F} = \sqrt[N]{F}\ln F$

$N = \ln F$ $\quad f = e$

~~$f = e$~~

⊙ $\gamma = 1 \rightarrow f = 4$

~~$\gamma + \sqrt[N]{e} - \frac{\sqrt[N]{e}}{N} = 0$~~

$N\gamma + N\sqrt[N]{e} - \sqrt[N]{e} = 0$

~~$N\gamma = \sqrt[N]{e}(1 -$~~

Questions (minor)
- ⊙ Switching threshold delay
- ⊙ delay problems
- ⊙ Gate sizing

Remembering
- ⊙ Mosfet characteristics

16/09/25 : ⊙ What is delay? Delay Fundamentals:

⊙ Important parameters to find delay?

Pg. no: 199 Kang

t_{PHL} – Time req for High to Low → Time req for i/p 50% & o/p 50% is t_{PHL}

t_{PLH} – Time req for Low to High → i/p 50%, & o/p 50% is t_{PLH}

$$\text{Delay}, t_p = \frac{t_{PHL} + t_{PLH}}{2}$$

⊙ How to calculate $V_{50\%}$?

$$V_{50\%} = V_{OL} + \frac{1}{2}(V_{OH} - V_{OL}) = \frac{1}{2}(V_{OL} + V_{OH})$$

$$V_{10\%} = V_{OL} + 0.1(V_{OH} - V_{OL})$$

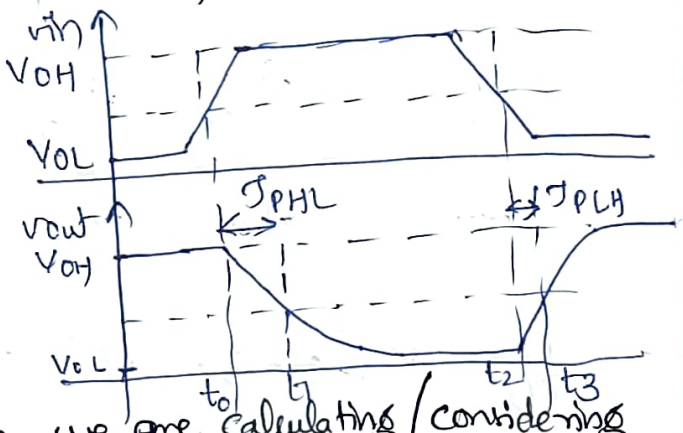
$$V_{90\%} = V_{OL} + 0.9(V_{OH} - V_{OL})$$

⊙ This helps to find the rise time and fall-time calculation.

⊙ Target is how to find, T_p?

$$i_c = C\frac{dv}{dt}$$

↑ mosfet current



⊙ Because of T_{PHL} – High to Low we are calculating/considering nmos. for pmos – T_{PLH} – low to High

$T_{PHL} = t_1 - t_0$

$T_{PLH} = t_3 - t_2$

⊙ i/p-High, o/p-High $V_{DS} > V_{GS} - V_T$ — nmos saturation.

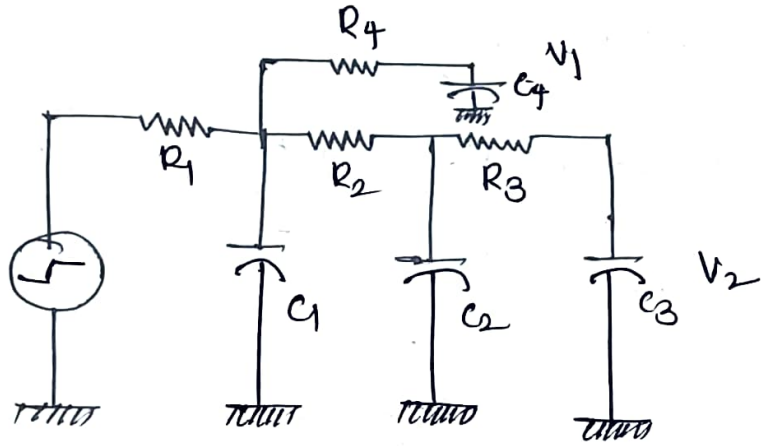
⊙ i/p High — ~~nmos~~ then capacitor will discharge voltage decreases gradually ie., High to low.

⊙ Until, $V_{DS} - V_T$ ⟶ nmos will be in saturation region less than that will be linear region.

$$\int_{t=t_0}^{t=t_1'} dt = -C_{load} \int_{V_{out}=V_{OH}}^{V_{out}=V_{OH}-V_{T,n}} \left(\frac{1}{i_{D,n}}\right) dV_{out}$$ ⟶ Saturation region.

$$\int_{t=t_1'}^{t=t_1} dt$$ ⟶ linear region.

⊙ Check how to get $t = 0.69RC$



$$T_{V_2} = R_1(C_1+C_2+C_3+C_4) + R_2(C_2+C_3) + R_3(C_3)$$

$$T_{V_1} = R_1(C_1+C_2+C_3+C_4) + R_4C_4.$$

→ Bcoz from R_1 we can see all capacitors C_1, C_2, C_3, C_4. then my path is towards V_1 so we go to R_4 then we see only C_4.

Charging : $V_{out} = V_{DD}(1 - e^{-t/RC})$

Discharging: $V_{out} = V_{DD}\, e^{-t/RC}$

⊙ As we know that delay is 50% ~~Vout~~ i.e., $0.5V_{DD}$

$$0.5 = 1 - e^{-t/RC}$$

$$0.5 = e^{-t/RC}$$

$$\frac{1}{2} = \frac{1}{e^{t/RC}}$$

$$2 = e^{t/RC}$$

$$t/RC = \ln 2$$

$$t = 0.69RC$$

30/09/2025: Check 195 page onwards.

Q) How to optimize the delay in series of inverters

C_{int} – internal capacitance.

C_{ext} – External capacitance

C_g – Gate capacitance.

f = fanout = $\frac{o/p}{input}$

γ – technology node.

F – Overall Fan out

N – No. of stages

f – fan out at a stage.

$$C_{int} = \gamma C_g$$

$$\frac{C_{g,j+1}}{C_{g,j}} = \frac{C_{g,j}}{C_{g,j-1}}$$

ie., optimum size is geometric mean of $C_{g,j-1}$ & $C_{g,j+1}$

198 page rabaey

$$C_{g,j} = \sqrt{C_{g,j-1}\, C_{g,j+1}}$$

$$f = \sqrt[N]{C_L/C_{g,1}} = \sqrt[N]{F}$$

minimum delay through the chain.

$$t_p = N t_{p_0}\left(1 + \sqrt[N]{F}/\gamma\right)$$

Note:
Our main aim is to Bring or find out fanout of all stages as same.

$f_1 = f_2 = \ldots = f_N$

$F = f_1 \ldots\ldots f_N$

$F = (f)^N$

$f = \sqrt[N]{F}$

$f = \sqrt[N]{C_L/C_{g,1}}$

$\gamma = 1, N = 1, F = 10$

$t_p = N t_{p_0}\left(1 + \sqrt[N]{F}/\gamma\right)$

$t_p = 11\, t_{p_0}$

$\gamma = 1, N = 2, F = 10$

$t_p = 2 t_{p_0}\left(1 + \sqrt[2]{10}\right)$

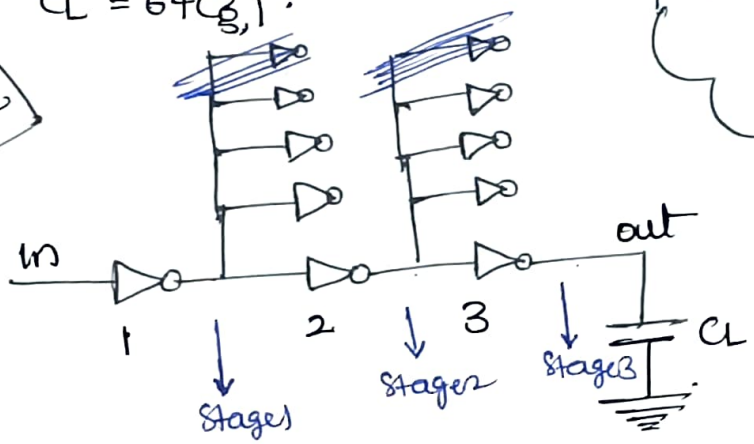
$t_p = 8.3\, t_{p_0}$

Q) Determine the sizes of the inverters such that the delay b/w nodes out and in is minimized. You may assume that

$C_L = 64 C_{g,1}$.

Here, only one inverter is driving 4 dps.

200 page



4 because inverter 1 is driving 4 dps so it is 4.

Overall Fan out

$$F = \frac{C_L}{C_{g,1}} \Rightarrow 64$$

$$f_1 = \frac{4C_{g,2}}{C_{g,1}}$$

$F = 64$

$$f = \sqrt[N]{F} = \sqrt[N]{64}$$

Here '2' is driving 4 dps.

$$f_2 = \frac{4C_{g,3}}{C_{g,2}}$$

$$f^N = 64 = 4^3$$

$$f_3 = \frac{C_L}{C_{g,3}}$$

$N = 3$

$$t_p = N t_{p0}\left(1 + \sqrt[N]{F}/f\right)$$

$$= 4 t_{p0}(1+4)$$

$$t_p = 20\, t_{p0}$$

$$\frac{4C_{g,2}}{C_{g,1}} = \frac{4C_{g,3}}{C_{g,2}} = \frac{C_L}{C_{g,3}}$$

Logical Effort (LE) 06/10/2021

↓

g

* For chain of complex gate

$$t_p = t_{p_0}\left(p + \frac{gf}{\gamma}\right)$$

t_{p_0} – intrinsic gate delay (unit)

γ – technology node.

f – electrical effort $= \frac{C_{ext}}{C_g}$

i.e., fan out

p = parasitic delay → Ratio of intrinsic delay of complex gate and simple inverter.

unitless but same as t_{p_0} (intrinsic gate delay)

↓

'p' value is coming from "layout" designing

$g = LE$ → logical effort: input capacitance of a gate by the input capacitance of a inverter for driving same o/p current.

$$\frac{\text{i/p capacitance of gate}}{\text{i/p capacitance of a inverter}}$$

Recap:

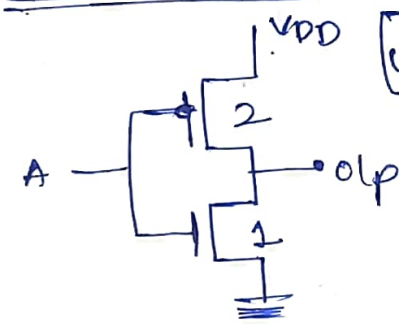
$$f = \sqrt[N]{F}$$

$$t_p = t_{p_0}\left(1 + \frac{C_{ext}}{C_g}\right)$$

$$t_p = t_{p_0}\left(1 + \frac{f}{\gamma}\right)$$

For chain of N inverter

Logical effort of a inverter:



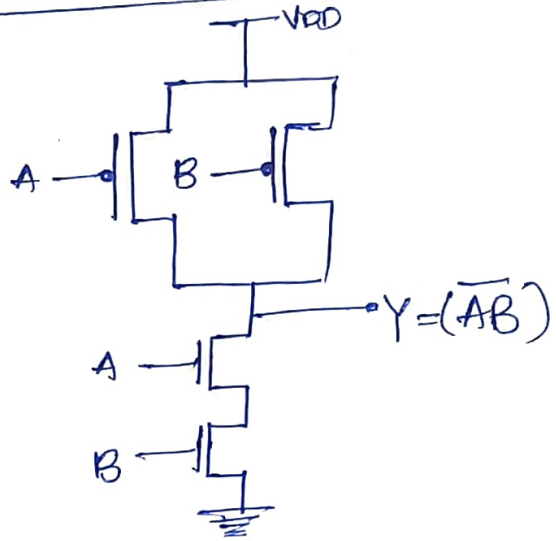
W ∝ C

i/p capacitance of a gate = 2+1 = 3

o/p capacitance of a inverter = 3

logical effort → $g = 3/3 = 1$

Find logical effort for Nand gate:



$x = 2$

$x = 2$

$x + x = 1$

$2x = 1$

$x = 1/2$

$\frac{1}{x} = \frac{1}{2}$

$x = 2$

$\frac{1}{x} + \frac{1}{x} = 1$

$\frac{2}{x} = 1$

$\frac{1}{x} + \frac{1}{x} = 1$

$x = 2$

$\frac{2}{x} = 1$

$x = 2$

2 i/p Nand gate : pmos – 2 nmos – 2

$$g = \frac{2+2}{3}$$

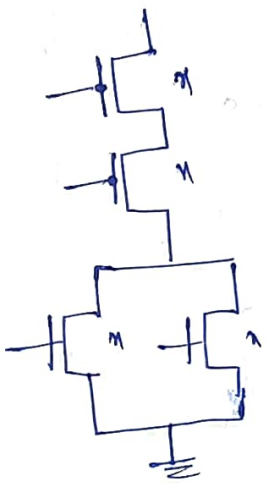
3 i/p nand gate : pmos – 2 nmos – 3

$$g = \frac{2+3}{3}$$

n i/p nand gate : pmos = 2 nmos – n

$$g = \frac{n+2}{3}$$

2 i/p Nor gate :



Pmos :

$\frac{1}{x} + \frac{1}{x} = \frac{1}{2}$

$\frac{2}{x} = \frac{1}{2}$

$x = 4$

$\frac{1}{x} = 1$

$x = 1$

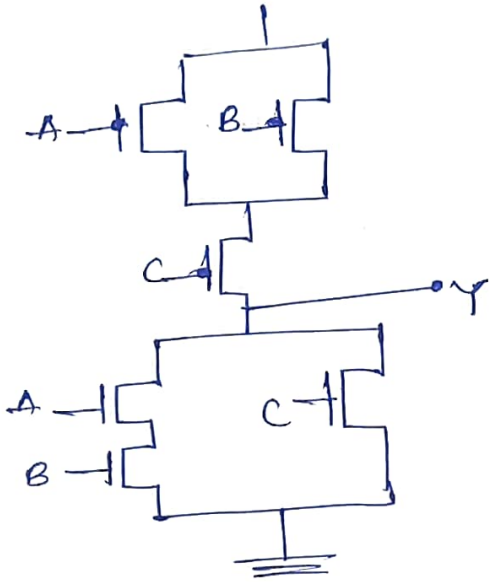
$g = \frac{5}{3}$

3 i/p = Pmos : 6 nmos = 1

$$g = \frac{6+1}{3}$$

n i/p Nor gate : $g = \frac{2N+1}{3}$

$Y = \overline{AB+C}$



Pmos

$\frac{1}{x} + \frac{1}{x} = \frac{1}{2}$

$x = 4$ $x_A = x_B = x_C = 4$

nmos

$\frac{1}{x} + \frac{1}{x} = 1$

$x = 2$ $x_A = x_B = 2$

$g = \frac{4+2}{3} = \frac{6}{3}$

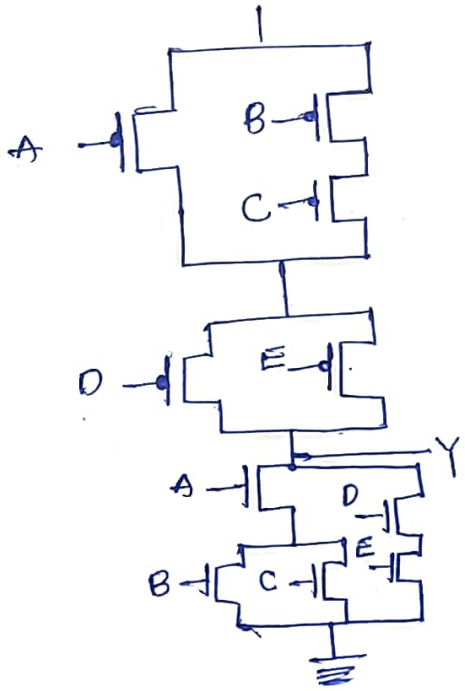
$\frac{1}{x} = 1$

$x = 1$ $x_C = 1$

$g_A = \frac{6}{3}$ $g_B = \frac{4+2}{3} =$

$g_C = \frac{5}{3}$

$Y = \overline{A(B+C) + DE}$



Pmos :

$\frac{1}{x} + \frac{1}{x} + \frac{1}{x} = \frac{1}{2}$

$x = 6$ — B, C, E, D

$\frac{1}{x} + \frac{1}{6} = \frac{1}{2}$

$\frac{1}{x} = \frac{1}{2} -$

$$\frac{1}{x} = \frac{3-1}{6}$$

$$\frac{1}{x} = \frac{x}{3}$$

$x=3$ — A

nmos:

$$\frac{1}{x} + \frac{1}{x} = 1$$

$x=2$ — A, B, C, D, E

$g_B, g_C, g_D, g_E = \frac{8}{3}$ → More capacitance more delay

$$g_A = \frac{5}{3}$$

⊙ So, we should not have more logical effort. Because if it is more then it has more delay.

Mixed CKS:

for 'N' no. of complex gates:

$$t_p = t_{p_0} \sum_{j=1}^{N} \left(p_j + \frac{f_j g_j}{\gamma} \right)$$

Delay (D)

F – overall Fan out

f – fan out

g – logical effort

G – overall logical effort

H – overall gate effort

h – gate effort

for optimal condn in series n – complex gate

$$f_1 g_1 = f_2 g_2 ----- = f_N g_N = h$$

$h = fg$ = gate effort

F = path effective fanout $\longrightarrow f_1 . f_2 . \dots f_N = \frac{C_1}{C g_1}$

G = path LE $= g_1 g_2 \dots g_N$

$$H = FG$$

$$h = \sqrt[N]{H} \quad \text{or} \quad h = \sqrt[N]{FG}$$

$H = f_1 f_2 \dots f_N \cdot g_1 g_2 \dots g_N$

$= (fg)^N \Rightarrow$

$h = fg = \sqrt[N]{H}$

$$h = \sqrt[N]{H}$$

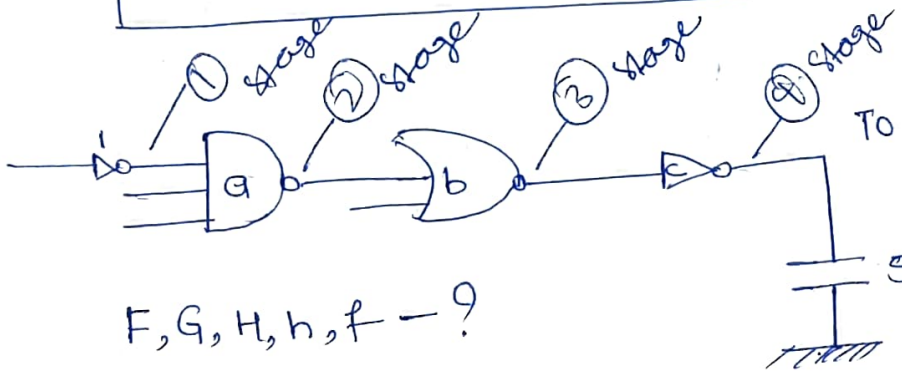
$$t_p = t_{p_0} \left(p_1 + p_2 + \dots + p_N + N . \frac{f_1 g_1}{\gamma} \right)$$

$$t_p = t_{p_0} \left(p_1 + p_2 + \dots + p_N + N . \frac{fg}{\gamma} \right)$$

$fg = h = \sqrt[N]{FG} = \sqrt[N]{H}$

$$t_p = t_{p_0}\left(\sum_{j=1}^{N} P_j + \frac{N\sqrt[N]{H}}{\gamma}\right)$$

Q)



To find P simply take no. of i/ps at the gate

$\gamma = 1$

F, G, H, h, f — ?

① $F = \frac{C_L}{C_{g_1}} = 5$

② $G = g_1 \cdot g_2 \cdots g_N$

$t_p = t_{p_0}\left(1+3+2+1+4\sqrt[4]{\frac{125}{9}}\right)$ $G = 1 \cdot \frac{5}{3} \cdot \frac{5}{3} \cdot 1 = \frac{25}{9}$

③ $H = FG = 5 \times \frac{25}{9} = \frac{125}{9}$

g_1 – inverter $= \frac{3}{3} = 1$

g_2 – Nand gate $= \frac{5}{3}$ (3 i/p)

g_3 – Nor gate $= \frac{5}{3}$ (2 i/p)

g_4 – inverter $= \frac{3}{3} = 1$

④ $h = \sqrt[N]{H} = \sqrt[4]{\frac{125}{9}}$

$= 1.93$

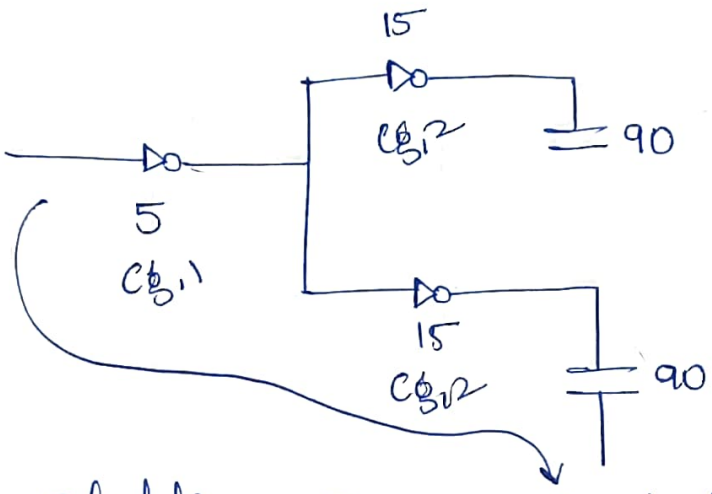
⑤ $f_1 g_1 = h \longrightarrow f_1 = \frac{1.93}{g_1} = \frac{1.93}{1} = 1.93$

$f_2 g_2 = h \longrightarrow f_2 = \frac{1.93}{5/3} = 1.158$

$f_3 g_3 = h \longrightarrow f_3 = \frac{1.93}{5/3} = 1.158$

$f_4 g_4 = h \longrightarrow f_4 = \frac{1.93}{1} = 1.93$

Branching factor:



Calculate → $F_1, G_1, H_1, h_1, \phi, f$

$$F = \frac{90}{5} = 18$$

$$G = 1 \cdot 1 = 1$$

$$H = FG_1 = 18$$

$$h = \sqrt[N]{H} = \sqrt{18} = 4.24$$

$$f_1 \phi_{S_1} = h \qquad f_2 \phi_{S_2} = h$$

$$f_1 = \frac{4.24}{1} \qquad f_2 = \frac{4.24}{1}$$

Not considered Branching factor so wrong solution

$$\boxed{H = GFB}$$

$$H = 1 \cdot 18 \cdot 2 = 36$$

$$h = \sqrt[N]{H} = \sqrt[2]{36} = 6$$

$$f_1 \phi_{S_1} = 6 \rightarrow f_1 = \frac{6}{1} = 6$$

$$f_2 \phi_{S_2} = 6 \rightarrow f_2 = \frac{6}{1} = 6$$

Note:
Our aim is to have same f for all stages so that we can have less delay

Ppt presentation of 10mins in DICD

Branching factor

$$f_1 = \frac{2 \cdot C_{g,2}}{C_{g,1}}$$

$$f_1 = \frac{15+15}{5}$$

$$f_1 = 6$$

$$f_2 = \frac{90}{15} = 6$$

$$f_1 f_2 \phi_{S_1} \phi_{S_2} = 6 \cdot 6 \cdot 1 \cdot 1 = 36$$

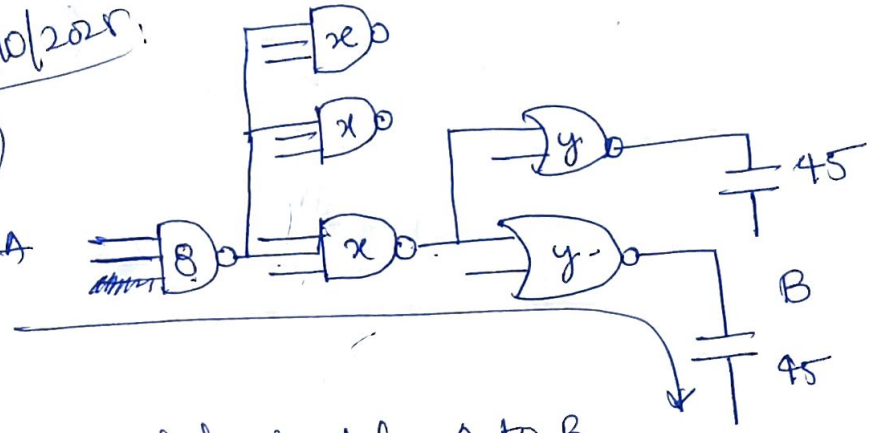
↓

different

∵ Bcoz of Branching factor

7/10/2025:

Q)



nand: $\frac{n+2}{3}$

nor: $\frac{2n+1}{3}$

Find min delay A to B,

$x, y = ?$

$$F = \frac{45}{8}$$

$$G = \frac{4}{3} \times \frac{5}{3} \times \frac{5}{3} = \frac{100}{27}$$

$$B = 3 \times 2 = 6$$

$$H = GFB$$

$$= \frac{45}{8} \times \frac{100}{27} \times 6 = 125$$

$$h = \sqrt[N]{H} = \sqrt[3]{125} = 5$$

assume $t_{p0} = 1$, $\tau = 1$

$$t_p = t_{p0}(2+3+2+3\times 5)$$

$$t_p = (7+15) = 22$$

$t_p = 22$

$f_1 g_1 = h$

$$f_1 = \frac{5}{4/3} = \frac{15}{4}$$

$f_3 g_3 = h$

$$f_3 = \frac{5}{5/3} = 3$$

$f_3 = \frac{4r \cdot 1r}{y} = 3$

$\boxed{y = 15}$

$f_2 g_2 = h$

$f_2 = \frac{5}{5/3} = 3$

$f_2 = 3 = \frac{2y}{x}$

$3 = \frac{2 \times 15}{x}$

$\boxed{x = 10}$

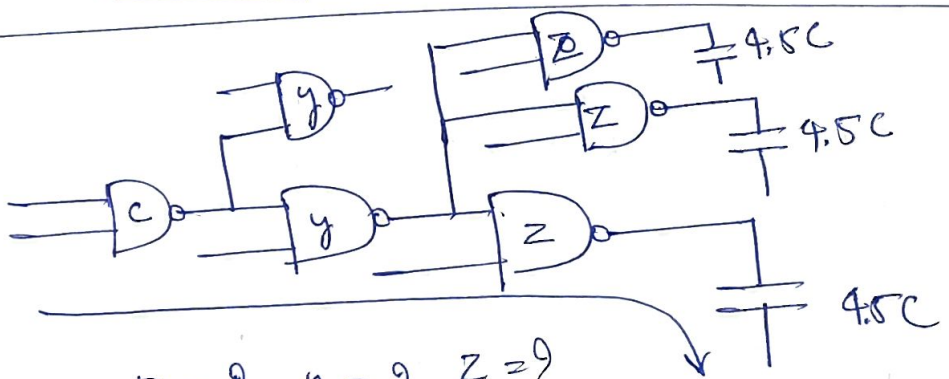
$f_1 g_1 = h$ verify : (recheck)

$f_1 = \frac{1r}{4} = \frac{3x}{z}$

$\frac{1r}{4} = \frac{3 \times 10}{z}$

$z = \frac{3 \times 10 \times 4}{15r}$

$\boxed{z = 8}$



D = ? y = ? Z = ?

$F = \frac{4.5C}{C} = 4.5$

$G = \frac{4}{3} \times \frac{4}{3} \times \frac{4}{3}$

$B = 2 \times 3 = 6$

$H = F G B$

$\leq 4.5 \times \frac{4}{3} \times \frac{4}{3} \times \frac{4}{3} \times 6$

$H = 64$

$h = \sqrt[N]{H} = 4$

$$t_p = t_{p0}(2+2+2+3\times4)$$

$$\boxed{t_p = 18}$$

$$f_3 g_3 = h$$

$$f_3 = \frac{4.5C\,4}{4/3}$$

$$f_3 = \frac{4.5C\,4}{4} \times 3$$

$$\frac{1.5C}{\frac{4.5C}{3}} = 3$$

$$\boxed{3 = 1.5C}$$

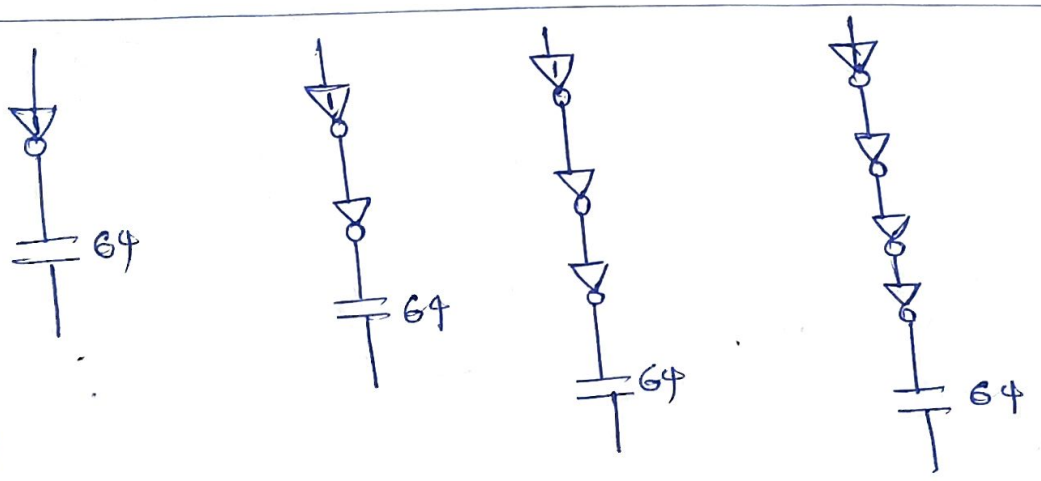
$$f_2 g_2 = h$$

$$f_2 = \frac{4}{4/3}\,3$$

$$f_2 = 3 = \frac{3Z}{y}$$

$$3 = \frac{3\times 1.5C}{y}$$

$$\boxed{y = 1.5C}$$



D?

$$t_{p_1} = (1 + 1\cdot\sqrt[1]{64})$$

$$t_{p_1} = 65$$

$$t_{p_2} = (2 + 2\sqrt[2]{64}) = 2(1+8) = 18$$

$$t_{p_3} = (3 + 3\sqrt[3]{64}) = 3(1+4) = 15$$

$$t_{p_4} = (4 + 4\sqrt[4]{64}) = \cancel{4(1+\ldots)} = 16$$

$$= \cancel{68}$$

(a) delay doesn't matter no. of stages. It is independent.

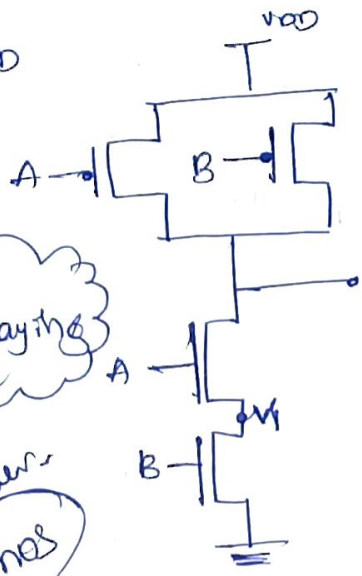
09/10/2021 :

"1" means V_{DD}

① A = B = 0 → 1

② A = 1, B = 0 → 1

③ B = 1, A = 0 → 1

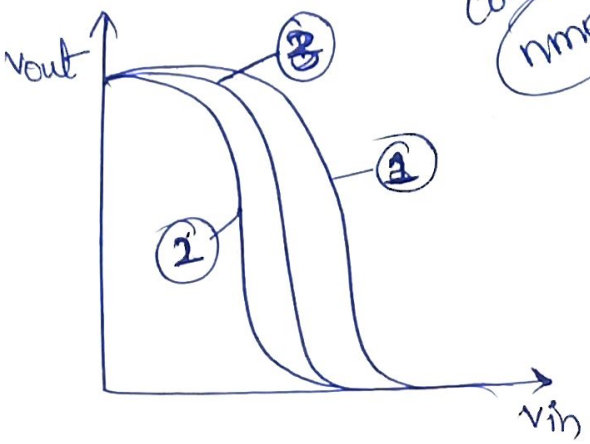


Why nmos? Because if it is 0 Pmos will be on Condn ie, Charging

Which is more delaying one

Draw VTC curve :

Consider nmos



① Consider resistance that will help to find the Vtc curve.

nmos

$V_{GS\,A} = V_A - V_1$

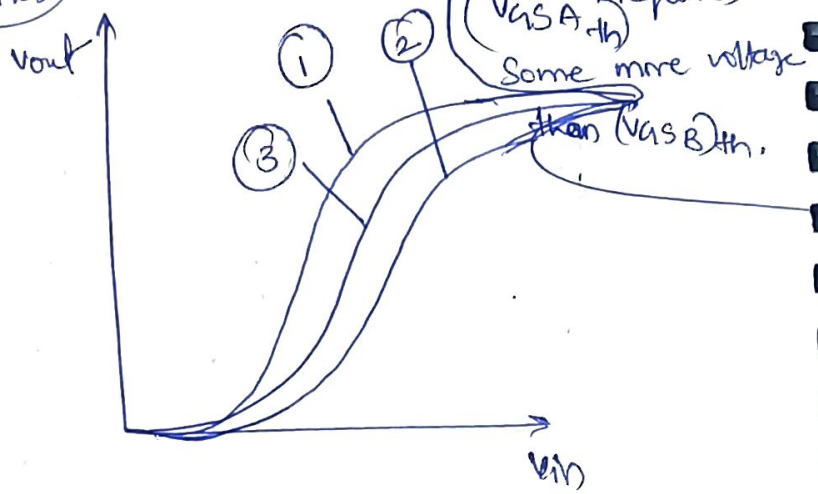
$V_{GS\,B} = V_B - 0$

① A = B = 1 → 0

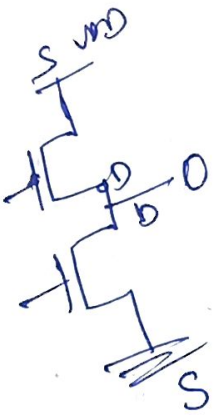
② A = 1, B = 1 → 0

③ B = 1, A = 0 → 1

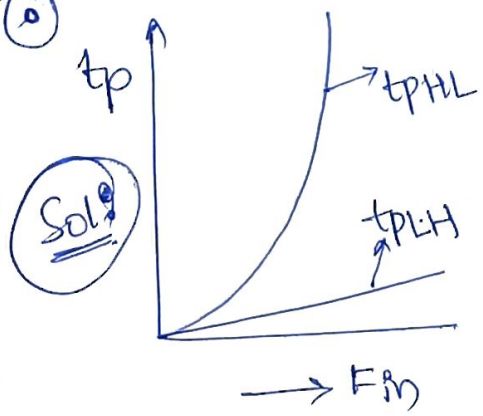
Pmos consider



Simulate nand in cadence so that u will know better



(1)



Sol:

t_{pLH} = pmos (charging) → Parallel ↑ resistance ↓ so decreases or constant

t_{pHL} = nmos (discharging) → Series ↓ so resistance increases. (NR)

Nand

Q) Draw tp vs Fin (No. of i/ps)

↓

(t_{pLH}, t_{pHL})

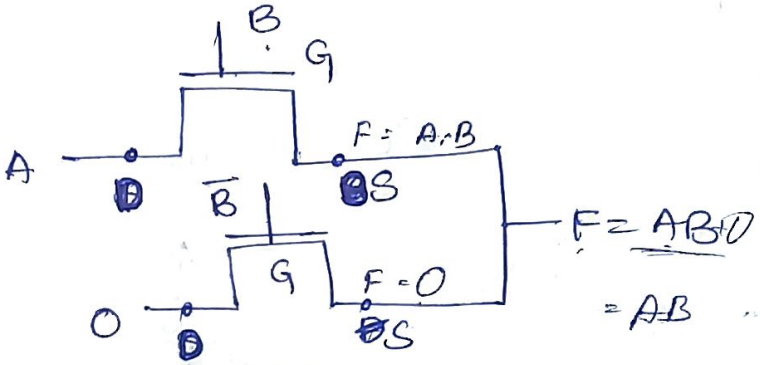
- Check search what lemore delay says? ooooo
- Go through Raebay & kang book.

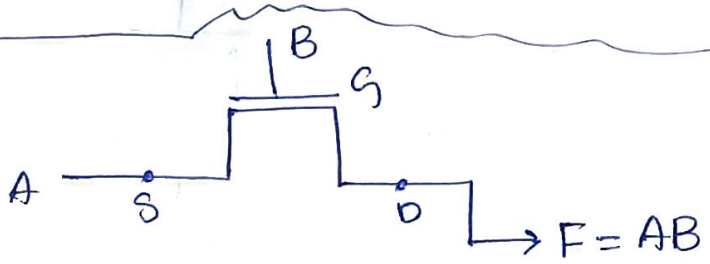
13/10/2021 : Pass Transistor : Mobility

NMOS based Pass Transistor

$F = AB$

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1





we cannot use this it wont work in 00 & 10 case beoz it will stay previous state only.

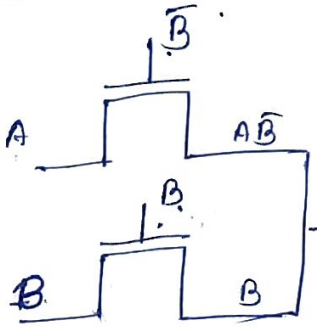
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

OR gate:

B → control i/p

B = 0 Y = A.

B = 1 Y = 1



$Y = A\bar{B} + B$

$= (A+B)$

$= B + AB$

$= (B+A)(B+B)$

$= (A+B) \cdot 1$

$= (B+A)(B+\bar{B})$

$= (B+A) \cdot 1$

$Y = A+B$

Design

$A + BC$

$A + BC$

$(A+B)(A+C)$

$AA + AB + BA + BC$

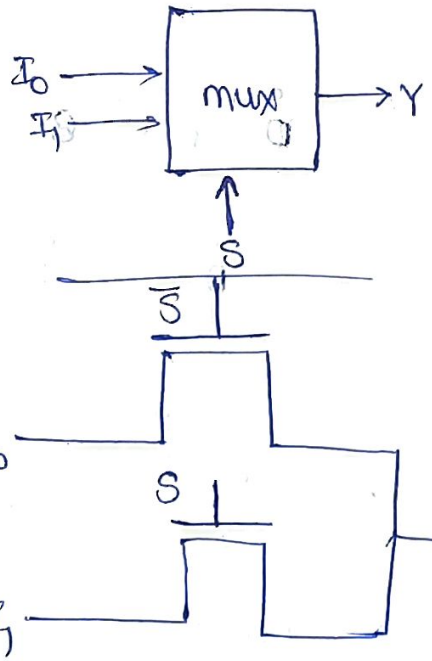
$A + AB + BA + BC$

$A + BC$

$A + BC = (A+B)(A+C)$

Design 2:1 mux:

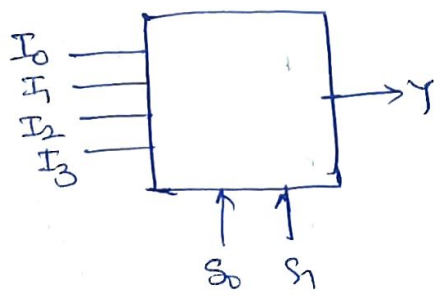
S	Y
0	I_0
1	I_1



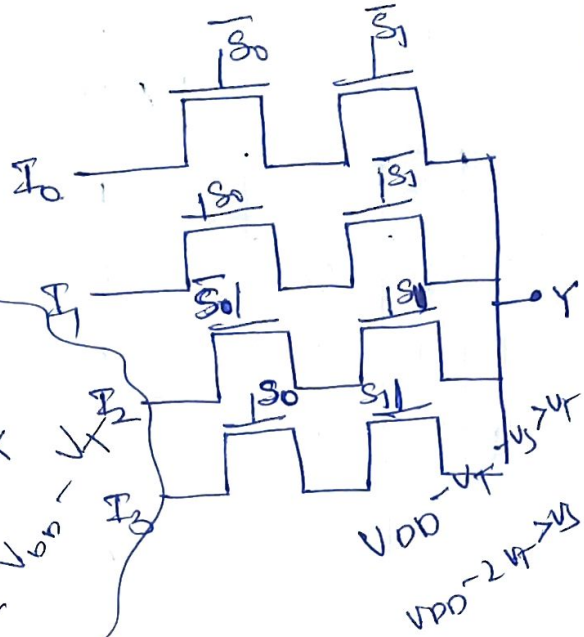
$Y = \bar{S}I_0 + SI_1$

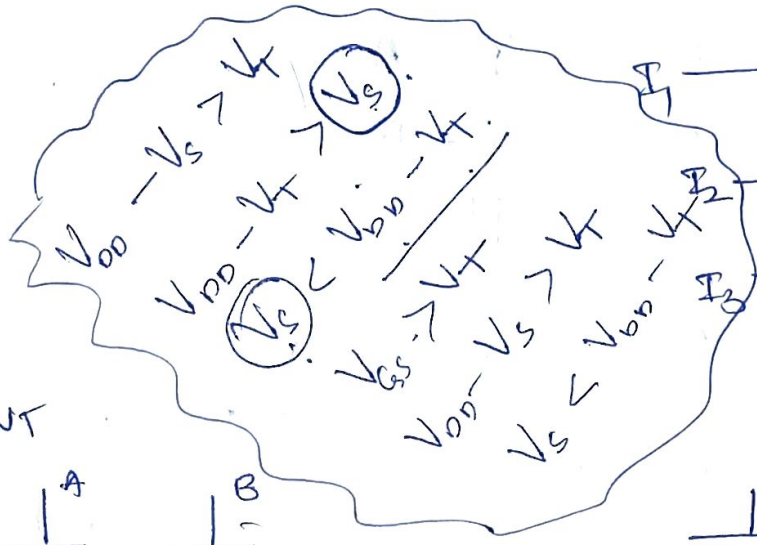
Try to do these designs & observe results in vivado.

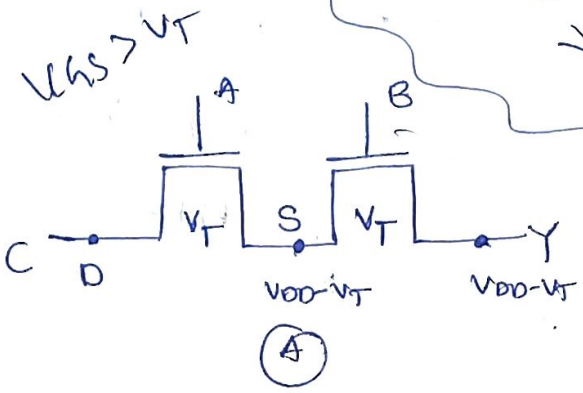
Design 4:1 mux :

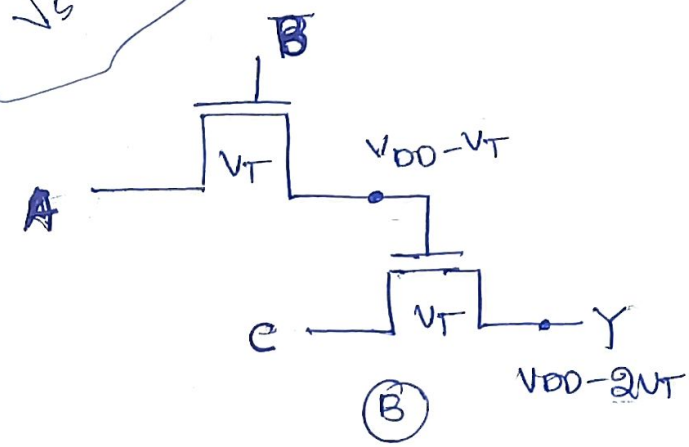


$$Y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$









Assume $A = B = C = V_{DD}$

A is good compared to B - voltage of A is $V_{DD} - V_T$

14/10/2021 : CPL (Complementary Pass Transistor logic)

① $A+B$ & $\overline{(A+B)}$

② AB & $\overline{(AB)}$

> DO combined $(A+B)$ & $\overline{(A+B)}$ then DO combinedly AB & $\overline{(AB)}$

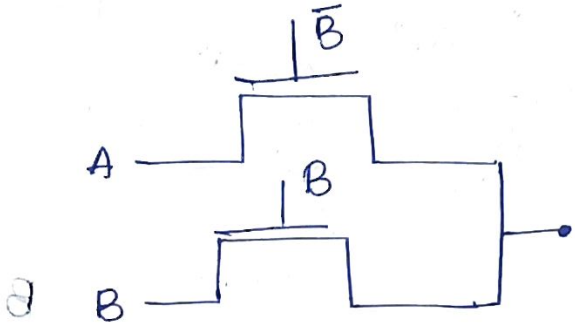
$A+B$ $\quad$ $\overline{A}\cdot\overline{B}$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

B – control i/p

B – 0 $\quad$ Y – A

B – 1 $\quad$ Y – B

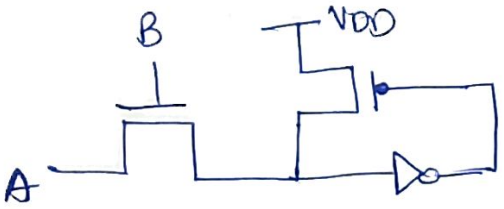


$Y = A\overline{B} + B$

$= (B+A)(B+\overline{B})$

$Y = B + B$

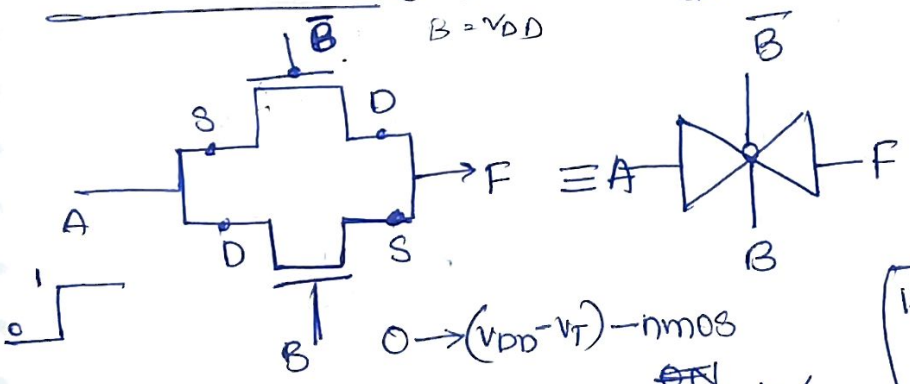
level restorer :



Search !

1. What is pass transistor?
2. What is level restorer?
3. What's the problem in pass transistor?

Transmission Gate (TG)



$0 \rightarrow (V_{DD} - V_T)$ – nmos ~~ON~~ working after that nmos is not working

Pmos is Saviour.

$V_{GS} < V_T$

$V_{DD} - V_T \rightarrow V_{DD} \rightarrow$ Pmos working.

$0 - V_{DD} < V_T$

$\therefore -V_{DD} < V_T$ → always on

$V_{GS} > V_T$

$V_{DD} - V_S > V_T$

$V_{DD} - V_T > V_S$

$V_S < V_{DD} - V_T$

In nmos :

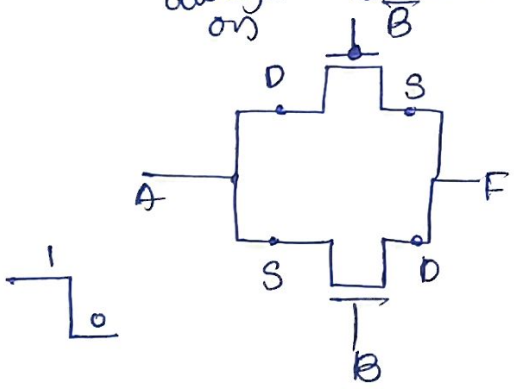
Higher potential is drain

lower " is source.

In pmos:

Higher potential is source

lower " is drain.



nmos: $V_{GS} > V_T$

$V_{DD} - 0 > V_T$

$V_{DD} > V_T$ always on

Pmos: $V_{GS} < V_T$

$V_g - V_S < V_T$

$0 - V_S < V_T$

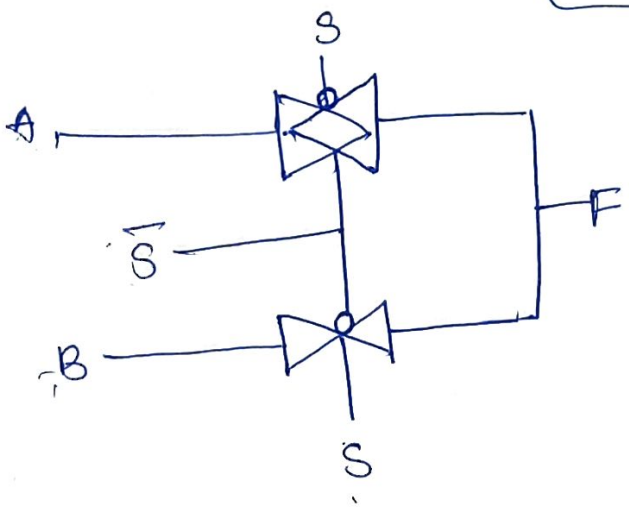
$\therefore -V_S < V_T$ work until ~~under~~ V_T

~~$V_S > -V_T$~~

⊙ Design TG based 2:1 mux:

For TG & Pass Transistor learning is good

$F = \bar{S}D_0 + SD_1$



⊙ Design TG Based XOR Gate:

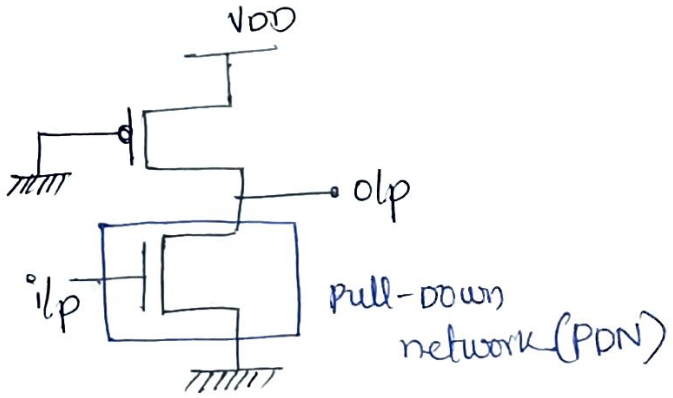
$Y = \bar{A}B + A\bar{B}$

⊙ Design XOR using 2:1 mux

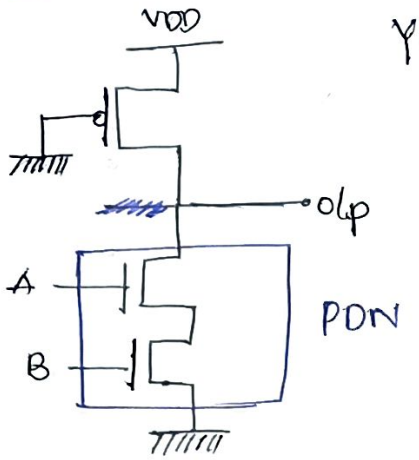
⊙ TG Based adder, XOR gate

16/10/2021 DICD: T-Spice:

Pseudo nmos:



Design nand gate using pseudo nmos:



$Y = \overline{AB}$

No. of Transistors in pseudo nmos	Cmos
N+1	2N

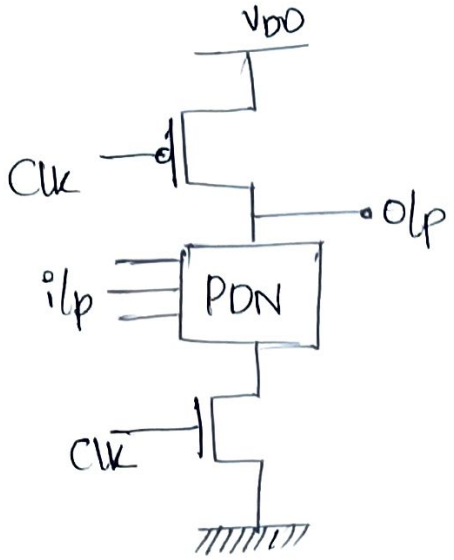
Note:
No. of Transistors are less in pseudo nmos then why we are not using this BCOZ
- Static power dissipation is more but in Cmos implementation any one of the transistor is ON. So zero power dissipatn.

⊙ Ratioed logic: W_{OL}, W_{OH}

olp of pseudo nmos depends w of pmos.

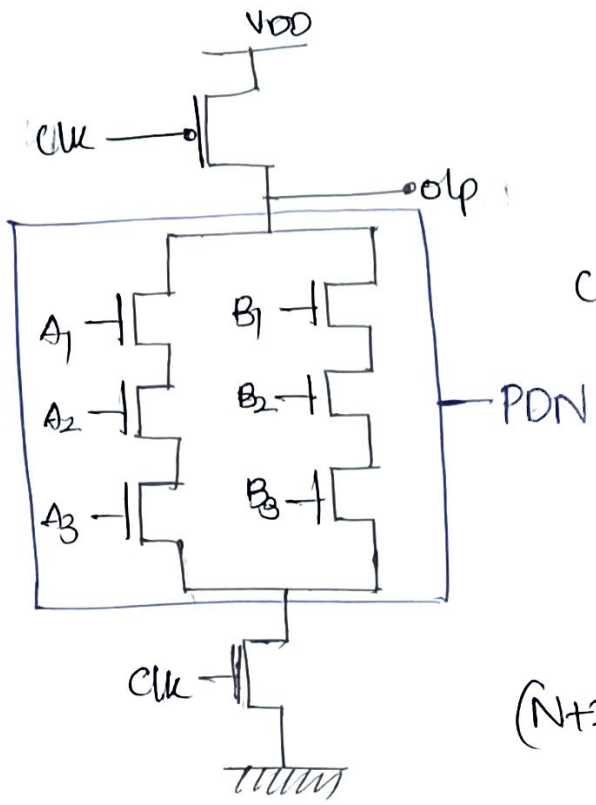
⊙ Non-ratioed logic (ratioless)

Dynamic logic : → means clock present

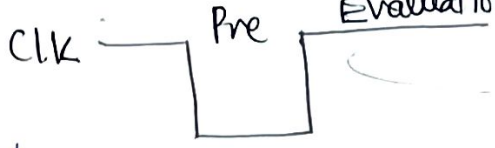


Important :
Physical design Engineer – Role is boom in S.C

$Y = \overline{A_1A_2A_3 + B_1B_2B_3}$ design using dynamic logic.



① Precharge ← when Clk = 0
② Evaluation. ← when Clk = 1

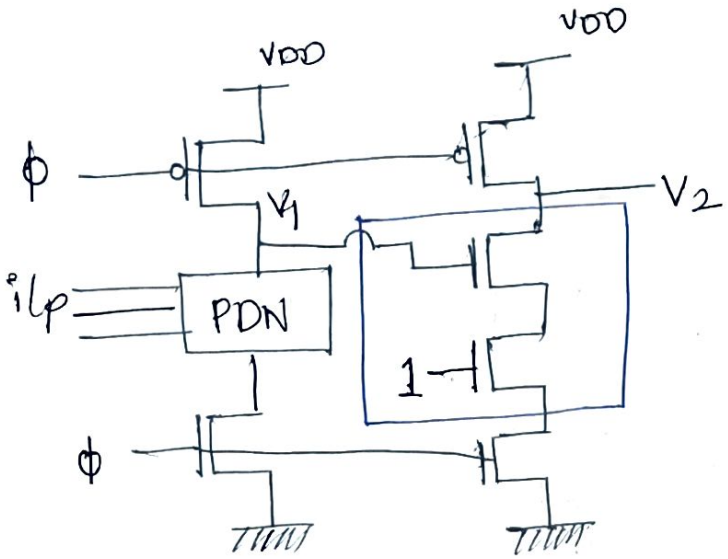


Static power dissipation — 0

(N+2) transistors.

No. of transistors is reducing then it is fth is reducing

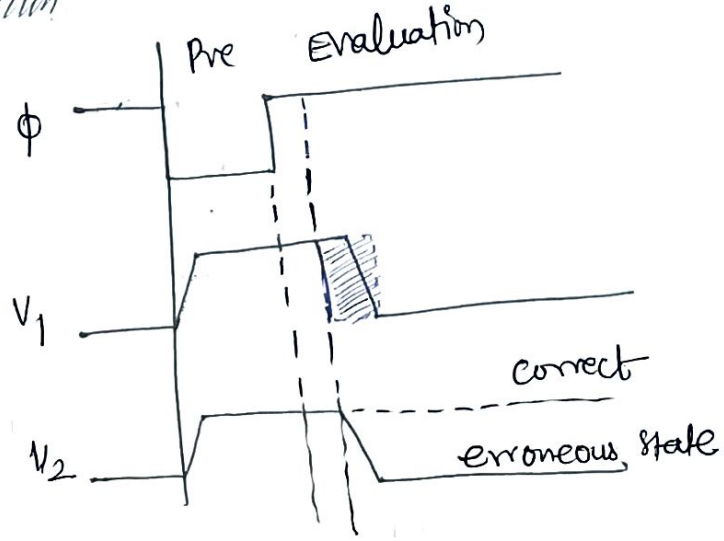
Kang (Dynamic logic problem): 399 page- NO



⊙ cascading two dynamic logic's is not good.

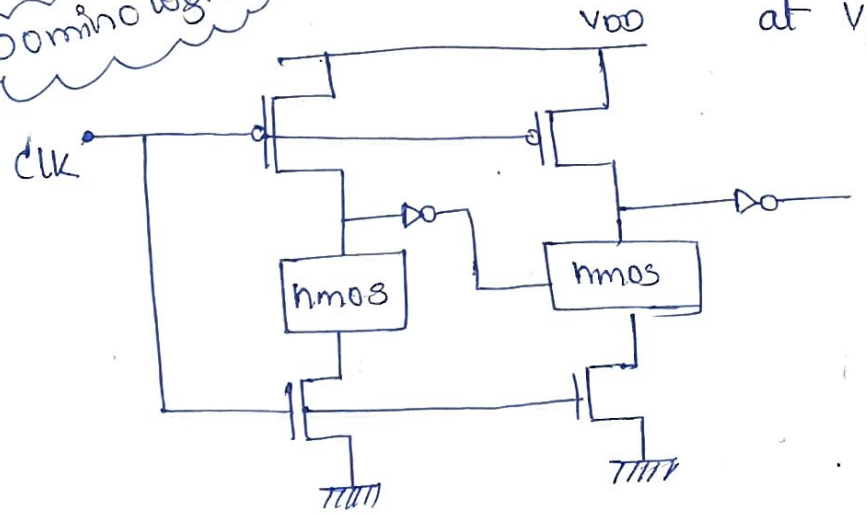
Erroneous: is caused due to delay happened in V_1.

NOTE: If there is no path to discharge then it will remain same as previous state.



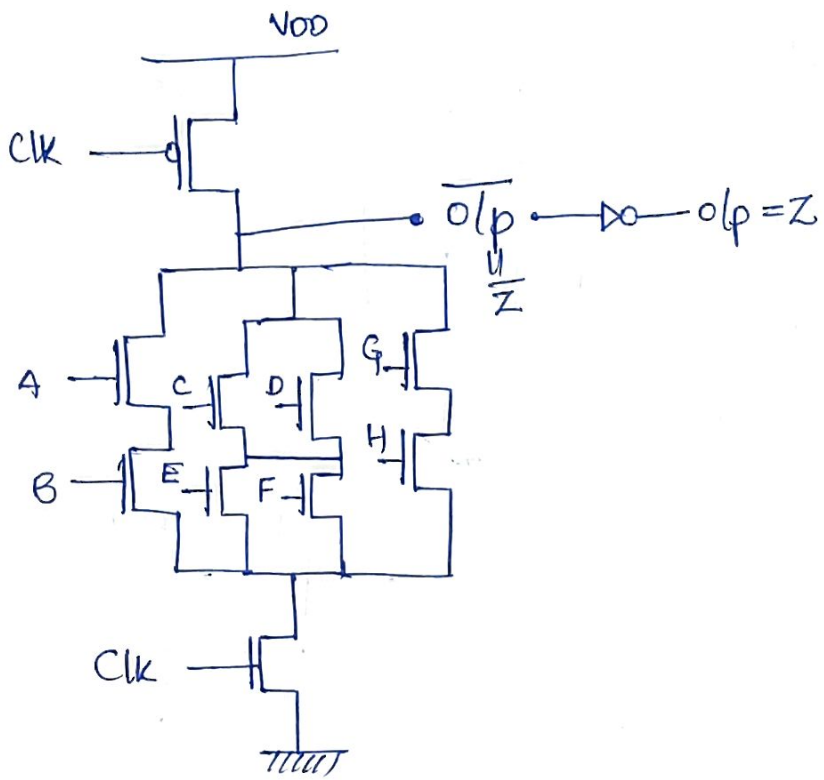
27/10/2025: Solu for dynamic logic problem: (Use inverter at V_1)

Domino logic



① Design using Dynamic logic

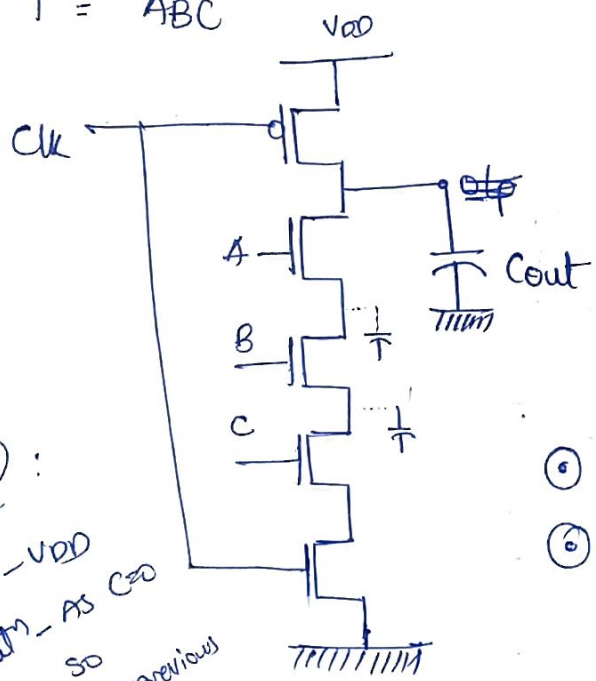
$$Z = AB + (C+D)(E+F) + GH$$



② Design 3 i/p Nand gate using Dynamic logic.

$$Y = \overline{ABC}$$

NAND3 → 3 i/p



→ ABC (0xx)

→ ABC (110)

ABC (110):

① Precharge – VDD

② Evaluation – As C=0 no path, so remains in previous state VDD.

ABC (0xx):

① During precharge V_{DD}

② " Evaluation A is 0 so no path, no discharge o/p remains in previous state

- In nmos capacitors b/w them there is internal node capacitors.
- In ABC(110) case, internal node capacitors will charge so it is the biggest problem i.e., charge leakage problem. o/p charge decreases.

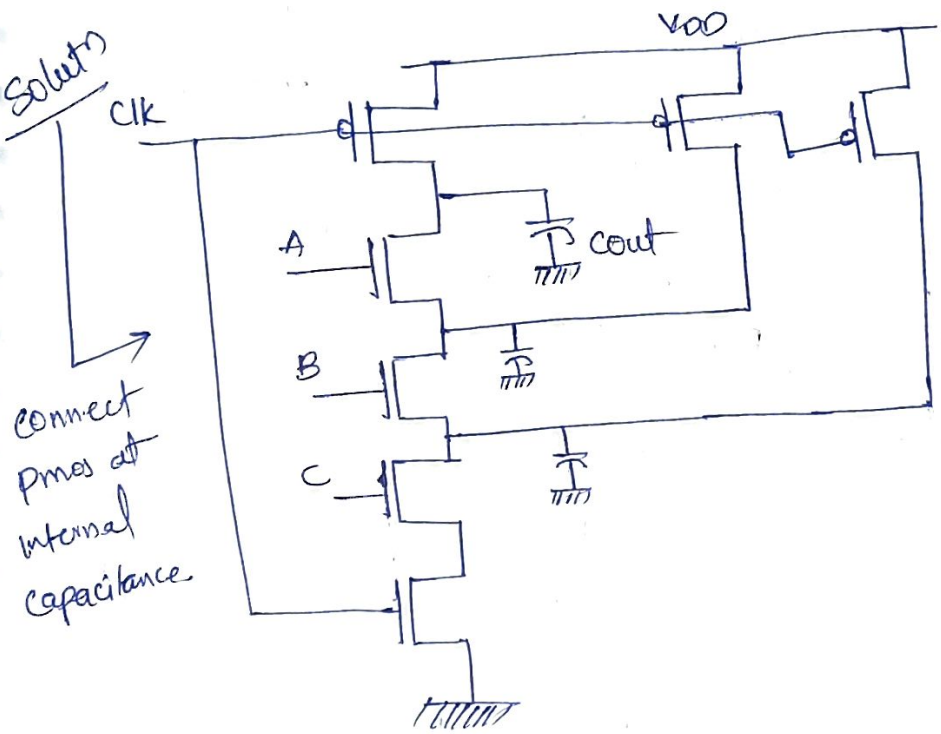
$$Q = C_{out} V_f + C_1 V_f + C_2 V_f = C_{out} V_{DD}$$

$$\boxed{V_f = \frac{C_{out}}{(C_{out}+C_1+C_2)} \cdot V_{DD}}$$

What is the solution for charge leakage?

- Cout can be high but bcoz of that delay, power dissipation will inc. so we can't.

Solution

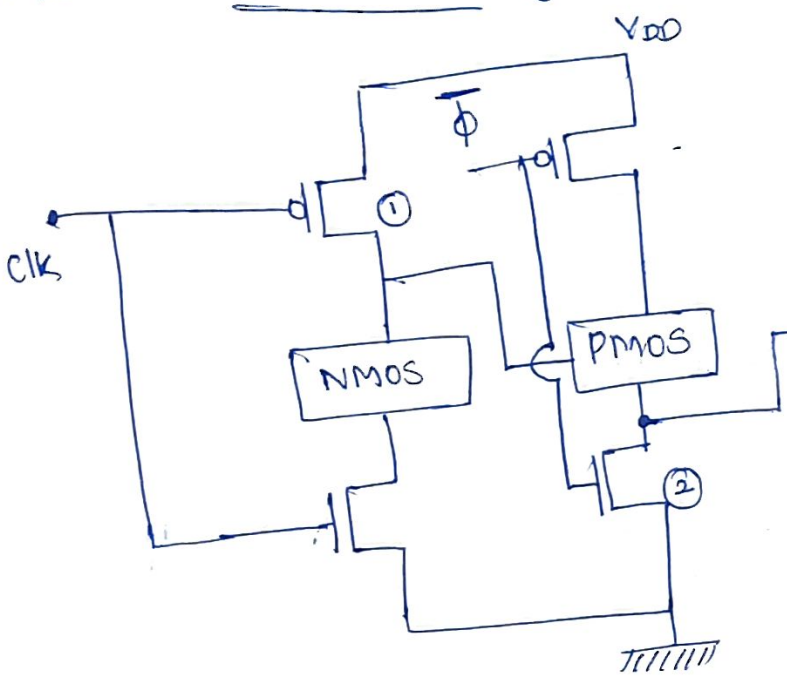


27/10/2021:

lect-2

NORA LOGIC : (C^2MOS – clocked CMOS)

↓ mainly used in pipelining architecture.



CLK = 0

① Precharge

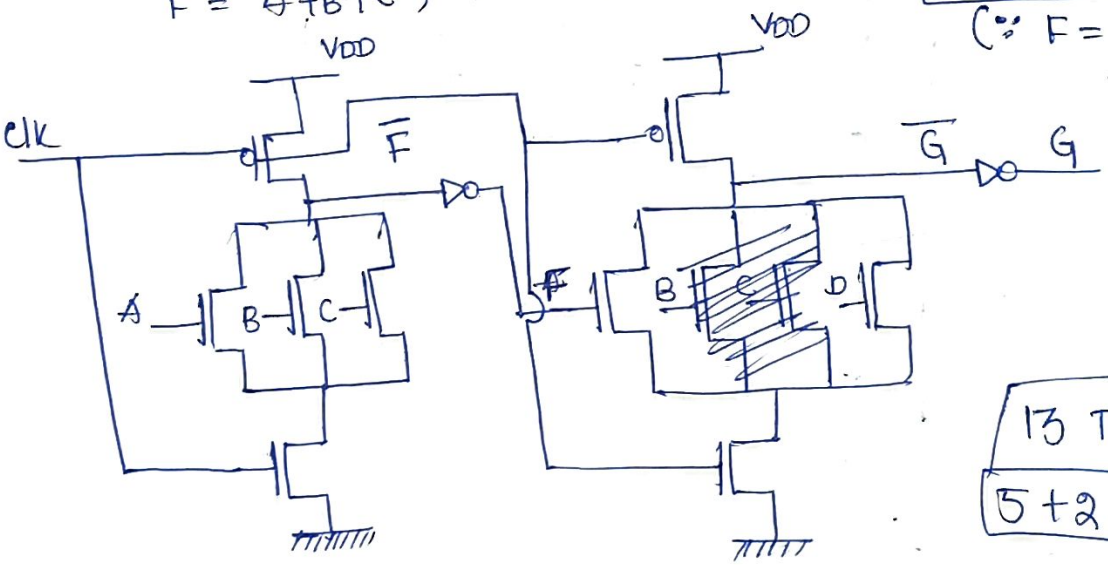
② Predischarge.

① Implement this function in dynamic cmos as cascaded with minimized no of transistors.

$F = A+B+C$, $G = A+B+C+D$

$G = F+D$

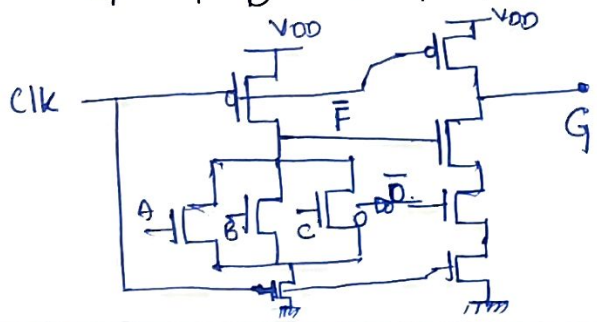
($\because F = A+B+C$)



13 Transistors.

5+2+4+2

$G = \overline{\overline{F}\,\overline{D}} = \overline{\overline{F}} + \overline{\overline{D}} = F+D$



11 Transistors.

Power:

$P_{dyn} = \alpha f C V_{dd}^2$ — f – clock frequency, α – switching factor.

(Need to reduce the α for dynamic power)

[No. of bits using also affects the α]

→ (i) Dynamic power (Switching)

(ii) Short circuit + power

(iii) leakage power (static power)

$$P_{avg} = \frac{1}{T}\left[\int_0^{T/2} V_{out}\left(-C\frac{dV_{out}}{dt}\right)dt + \int_{T/2}^{T}(V_{DD}-V_{out})\left(C\frac{dV_{out}}{dt}\right)dt\right]$$

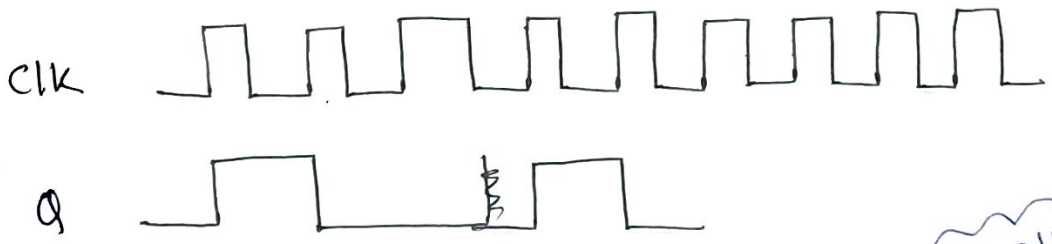
In dynamic logic we get dynamic power which is very high. ie., Bcoz of using clk.

28/10/2021:

$$\alpha_{0\to1} = \frac{N_0}{2^N} \times \frac{N_1}{2^N}$$

$\alpha = \frac{2}{8} = 25\%$

$\alpha = ?$



$$= \frac{N_0(2^N - N_0)}{2^{2N}}$$

Kevin Rabaey book

q → for NOR gate

$q_{0\to1} = \frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

$\frac{1}{4}$: $\frac{\text{No. of ones}}{\text{Total}}$

$\frac{3}{4}$; $\frac{\text{No. of zero's}}{\text{Total}}$

q → for XOR gate :

$0 q_{\to1} = \frac{2}{4} \times \frac{2}{4} = \frac{4}{16}$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

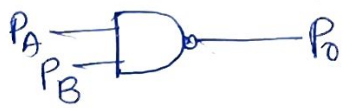
q → for 3 i/p NOR gate :

$q_{0\to1} = \frac{1}{8} \times \frac{7}{8} = \frac{7}{64}$

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Generalized form :

Two i/p NAND gate mutually independent prob. P_A P_B



$P_0 = P_A P_B$

$P_1 = 1 - P_0 \Rightarrow 1 - P_A P_B$

~~P_A P_B~~

① Two i/p NOR gate

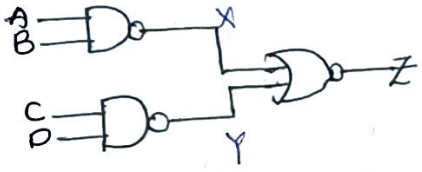


$$P_1 = \overline{P_A + P_B} = \overline{P_A} \cdot \overline{P_B}$$
$$= (1-P_A)(1-P_B)$$

$$P_0 = 1 - \overline{P_A} \cdot \overline{P_B}$$

② Find the dynamic power of the ckt

$$P_A = P_B = P_C = P_D = 0.5$$



$$P_{X_0} = P_A P_B = 0.5 \times 0.5 = 0.25$$

$$P_{X_1} = 1 - P_A P_B = 1 - 0.5 \times 0.5$$
$$= 0.75$$

$$\alpha_X = P_{X0} \cdot P_{X_1} = 0.25 \times 0.75$$

$$\alpha_Y = P_{Y_0} \cdot P_{Y_1} = 0.25 \times 0.75$$

$$P_{Y_0} = P_C P_D = 0.5 \times 0.5 = 0.25$$

$$P_{Y_1} = P - P_C P_D = 1 - 0.5 \times 0.5$$
$$= 0.75$$

$$P_{Z_1} = \overline{P_{X_0}} \, \overline{P_{Y_0}}$$
$$= (1 - P_{X_0})(1 - P_{Y_0})$$
$$= (1-0.25)(1-0.25)$$
$$= 0.75 \times 0.75$$
$$= 0.5625$$
$$= 0.0625$$

$$P_{Z_0} = 1 - P_{Z_1}$$
$$= 1 - 0.0625$$
$$P_{Z_0} = 0.9375$$

$$\alpha_Z = P_{Z_0} \cdot P_{Z_1} = 0.9375 \times 0.0625$$

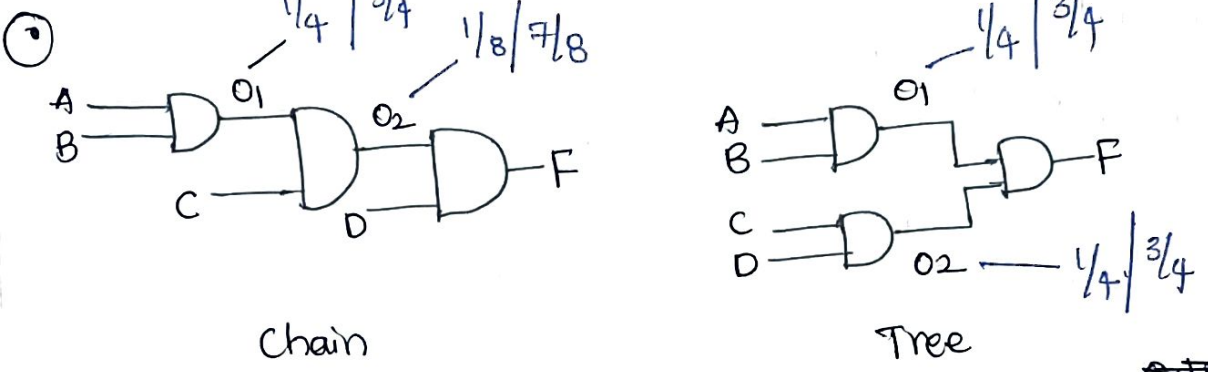
$$\alpha_Z = 0.058$$

dynamic power $= \alpha f C V_{dd}^2$

$= \alpha_x f C_x V_{dd}^2 + \alpha_y f C_y V_{dd}^2$

$+ \alpha_z f C_z V_{dd}^2$

$P_{dyn} = (\alpha_x C_x + \alpha_y C_y + \alpha_z C_z) f V_{dd}^2$



Chain Tree

We get same o/p for both Chain & Tree then which is best?

② α of Chain is good (lower) compared to tree.

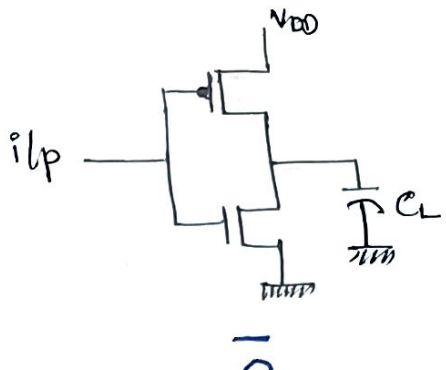
① ① Tree structure is good in terms of delay ② glitch ↓ unwanted spikes

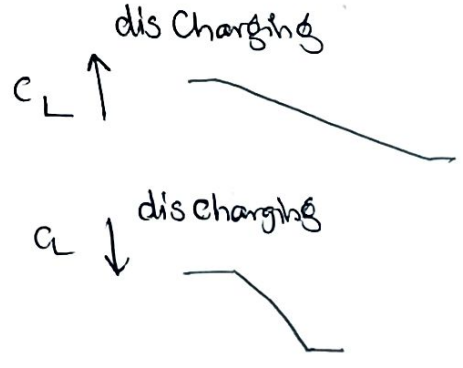
② overall α is same for both Chain & Tree.

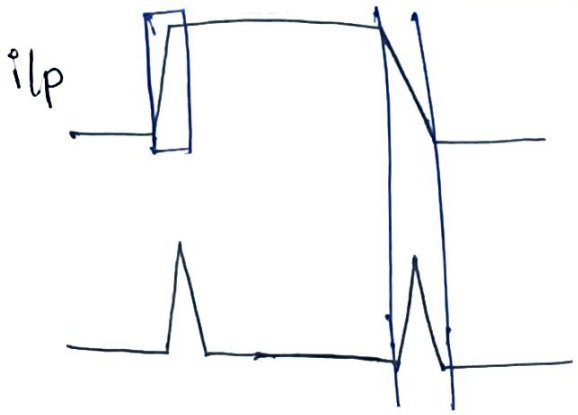
then why still we use tree?

29/10/2021:

Short ckt power:







(i) Leakage – Sub-threshold leakage power

– Gate to Drain tunneling (band-to band tunneling)

$C_{ox} = \frac{\varepsilon}{d}$ $C_{ox} \downarrow, I \downarrow$ (BCOZ $I \propto \mu$ for W)

Q) How we can reduce the overall power?

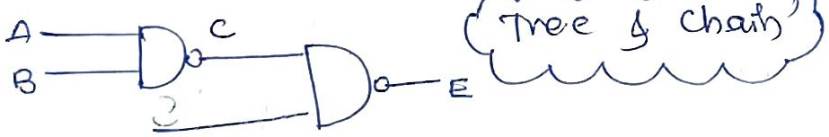
⊙ reduce Vdd

⊙ Clock gating technique.

⊙ Scaling (S) – reducing the size of the Transistor

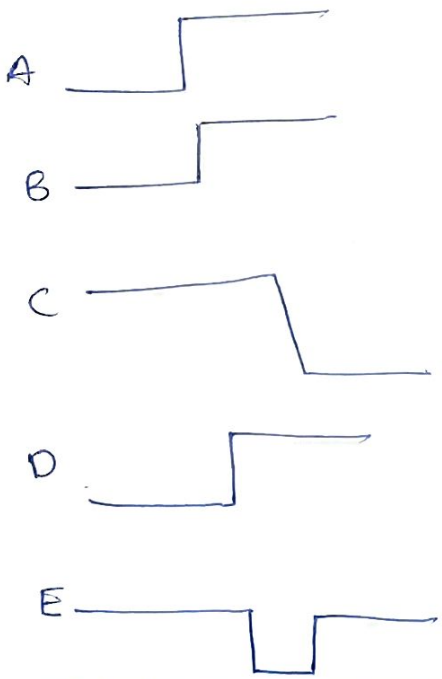
Constant field scaling (CFS) → constant voltage scaling (CVS) →

Normal simulation
postout simulation
↓
In this we can delay in the o/p but we can't see in Normal simulation.

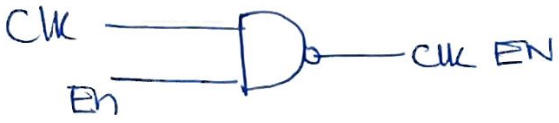


Chain structure

⊙ Use Balanced based structure to avoid delay, glitches.
ie. Tree Structure



Clock gating



⊙ When clock is req, Enable the bit

Scaling (S):

(CFS) (CVS)

$W = W/S$

$L = L/S$

$t_{ox} = \frac{t_{ox}}{S}$, $V_{DD} = V_{DD}/S$, Junction thickness (depth of drain or source)

$V_T = V_T/S$ (threshold voltage)

$I = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{DS})^2$

$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{S\varepsilon_{ox}}{t_{ox}}$

$I = \mu_n \frac{S\varepsilon_{ox}}{t_{ox}} \frac{W/S}{L/S} \left(\frac{V_{GS} - V_{DS}}{S}\right)^2$

$I \propto S/S^2$ $I_{new} \propto \frac{1}{S}$

$V_{new} = \frac{V_{old}}{S}$

constant voltage (V_T & V_{DD})

$I \propto S$

$V = V$

$P = IV$

$P = SIV$

$P \propto S$

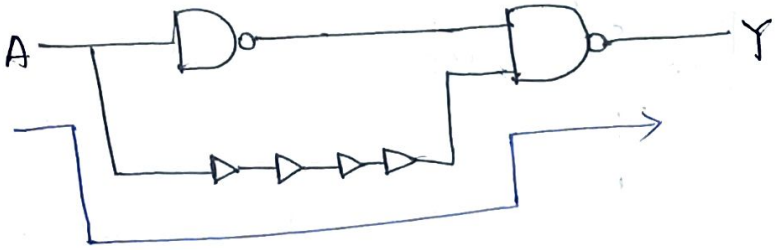
$P = VI$

$= \frac{I_{old}}{S} \cdot \frac{V_{old}}{S}$

$P = \frac{V_{old} I_{old}}{S^2}$

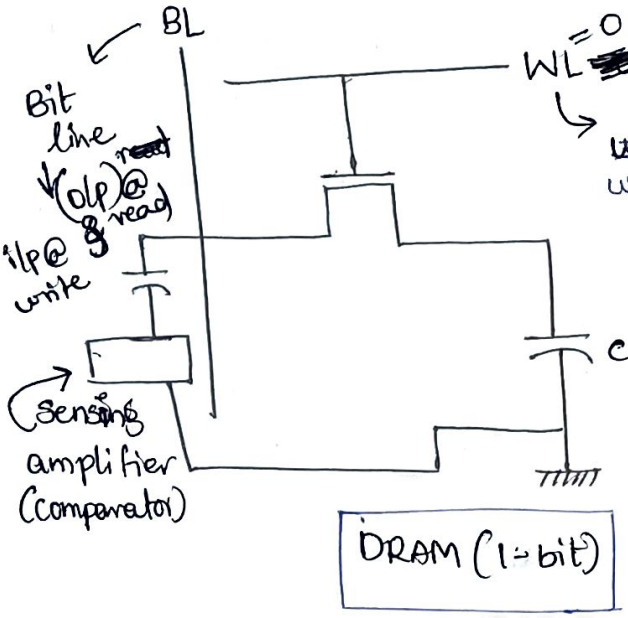
CFS $P \propto \frac{1}{S^2}$

multiple V_T (controlled by gate substrate voltage)



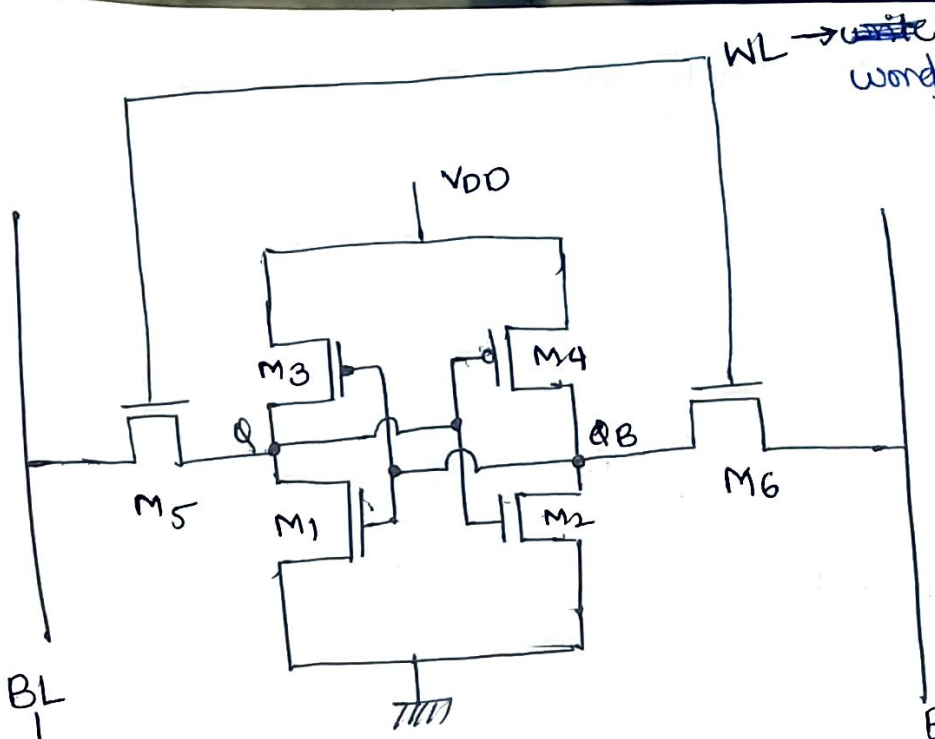
What is critical path.

$$f = \frac{1}{\text{which takes longer path } (t_{critical\ path})}$$



memory < DRAM, SRAM

- Bit line — acts as o/p @ read & i/p @ write.
- Comparator is used as sensing amplifier.
- problem is leakage from capacitor. we should refresh every time the capacitor ie., charge everytime.

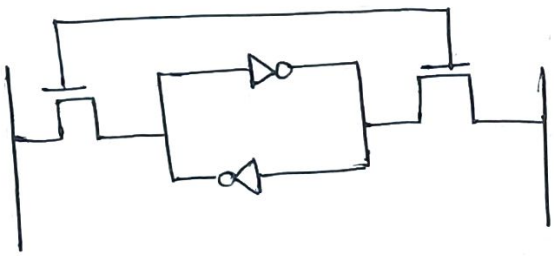


Q is o/p of M_3 & M_1

Q is i/p of (M_4 & M_2)

Q_B is o/p of (M_4 & M_2) & i/p of M_3 & M_1.

SRAM (1 unit of SRAM cell)



BL & BLB — are two distinct lines. not related to eachother.

Cases:

① WL = 0 → Hold forcefully. M_5 → OFF ✗ M_6 → OFF ✗

② WL = 1 → Conditions: Q = 1, Q_B = 0, BL = 0, BLB = 1

M_5 ✓ M_6 ✓

want to write = 0 at Q.

Consider mostly Q BCoz Q_B is opposite of Q.

from Q = 1 to BL = 0 discharge

So Q will become 0.

③ WL = 1 → Conditions.: Q = 0, Q_B = 1, BL = 1, BLB = 0

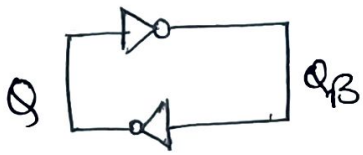
want to write = 1 at Q

WL = 1 ; — otw it won't work — (m_5, m_6) only in Hold Condn WL = 0

want to read 0 → Think it as DRAM.

BL = 1, BLB = 1 , Q = 0, Q_B = 1

acts as reading.



Q = 0 ie., o/p of M_3 & M_1 then

ie., Q = 0 is i/p of M_2 & M_4

then M_4 is ON & M_2 is OFF

then Q_B is ON. Then M_3 is OFF & M_1 is ON.

So From BL = 1 to M_1 will discharge to gnd.

4/11/2025 :

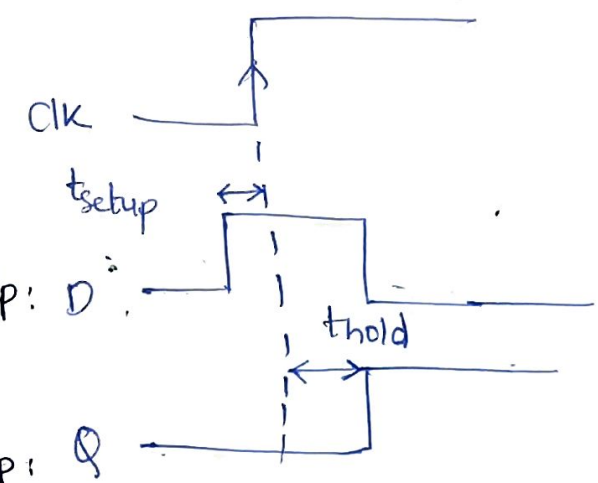
Sequencial Logic :

VLSI

~~Used~~ in CAD – computer Aided design.

STA (Static Time Analysis)

- Setup time
- Hold time



- at t_{setup} time, i/p should not change.
- at t_{hold} time, o/p should not change.

Check what is setup time & Hold time ?

Latch & Flip flop : — Storage element

- Level triggered
- edge Triggered.

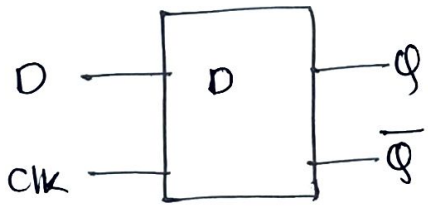
PPT:

Floating point numbers.

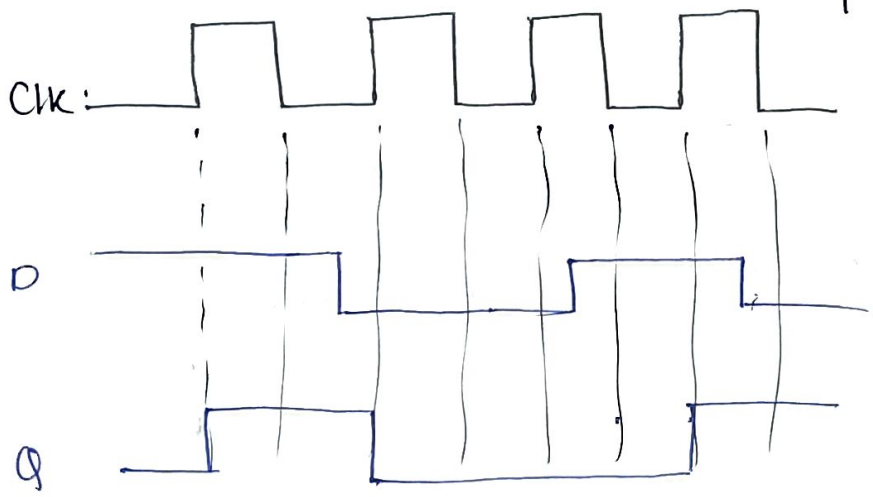
IEEE 745

- NaN
- SgnNan

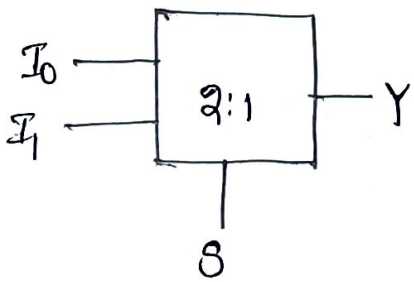
D-Latch with clk:



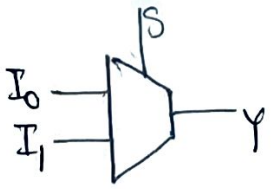
CLK	D	o/p
0	x	Q_n
0	0	Q_n
1	1	



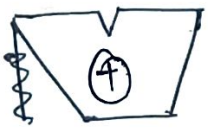
mux-Based latch:



S	Y
0	I_0
1	I_1



mux



Adder

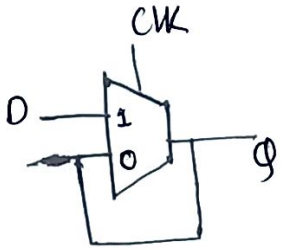


multiplier

$\gg$ $<<$

Shift registers

ie,



$\Longrightarrow$ Mux Based Latch.

(mux acts as combinational & also sequential)

Draw timing diagram:

